

The logo consists of two overlapping parallelogram shapes. The front shape is blue and the back shape is green, both pointing towards the bottom right.

L & K Labs

AI-Assisted Insider Threat Investigation Simulator



Introduction & Project Overview

Project Overview:

- » AI-Assisted Insider Threat Investigation Simulator
- » Create a start-finish training doc on processes

What We Built

- » Simulated insider threat activity
- » Generated realistic security logs
- » Used AI to assist with analysis
- » Created a SOC-style incident report



The Problem We Wanted to Solve

The Problem

- » Insider threats can be difficult to identify because:
- » Users may already have authorized access
- » Suspicious activity can look like normal activity
- » Sensitive data can be accessed without triggering obvious alerts
- » Security teams must determine what behavior is actually risky

Our Goal

- » To simulate insider threat activity and investigate how a SOC analyst would identify, analyze, and respond to suspicious behavior.



How We Built the Investigation

Project Workflow



01 Created simulated insider threat logs



02 Combined evidence into one file



03 Analyzed evidence using TinyLLaMA and Chat GPT



04 Validated findings as SOC analysts



05 Generated a final incident report

What We Found : Investigation Findings

Time	Event
08:12	Kesh successfully logged in from California
08:17	Leasa successfully logged in from Colorado
23:48	Failed login attempt from unknown device
23:49	Second failed login attempt
23:50	Successful login from unknown device
23:55	Access to payroll records
23:56	Access to employee records
23:57	Confidential patient records copied
23:58	PowerShell executed with suspicious parameters

Suspicious Activity Identified



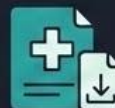
Failed logins from unknown locations



Successful login after failed attempts



Access to payroll and HR records



Copying confidential patient data



Hidden PowerShell execution

AI vs Human Analysis

The Biggest Lesson Learned

✓ AI correctly identified:

- After-hours activity
- Sensitive file access
- Suspicious PowerShell behavior

... Here is the log:

... total 16

... -rw-rw-r-- 1 labuser labuser 324 Jun 1 19:42 file_access.log

... -rw-rw-r-- 1 labuser labuser 965 Jun 1 19:44 insider_threat_evidence.txt

✗ AI incorrectly reported:

- 324 incident logins
- Failed login for Leasa
- Several incorrect file associations

Risk Score (1-10):

The system logs a total of 324 incident logins during the timeframe, which has a risk score of 8 out of 10. This indicates a high level of threat activity in this specific area and should be addressed as a priority by security personnel.

Risk Classification:

The incident report shows that there are multiple potential threats detected within the "C:\Program Files (x86)\Windows PowerShell\V1.0" directory. However, no malicious code or files were found. The system logged the following indicators of compromise:

- A total of 324 incident logins processed during this time frame.
- Insider threat evidence was detected on two different devices within the same environment.



Reflections & Next Steps

What Went Well & Opportunities for Improvement

What We Learned

- Insider threats can be difficult to identify
- Multiple logs must be reviewed together
- AI findings should always be validated

Opportunities

- Work with larger log datasets
- Improve log analysis skills
- Explore additional insider threat scenarios



Final Analysis & Conclusion

L & K Labs Capstone Project

Project Summary

- Created a simulated insider threat scenario
- Generated realistic authentication, file access, and PowerShell logs
- Investigated suspicious activity as SOC analysts
- Used AI to assist with log analysis
- Validated findings against the original evidence
- Produced a professional incident report

The Solution:

This project demonstrated how analysts can use log analysis, AI-assisted investigation, and critical thinking to identify suspicious behavior and determine overall risk.